

Final Term

OBE

1. System of Linear Equations :

- Gaussian Elimination + (with partial pivoting)
- Jacobi method
- Gauss-Seidel method

2. Non-linear Equation in one variable :

→ 3 questions

- Graphical method
- Bisection method
- Secant method
- Newton's method
- fixed point iteration

3. Non-linear Equation in ~~two~~ _{more} variables :

- fixed point
- Newton's method

4. polynomial fitting / interpolation :

- Lagrangian
- Newton's dividend method.

+2
+1
3 questions

5. spline

- linear
- cubic

- Natural cubic spline.
- Clamped cubic spline.
- knot-a-knot cubic spline.

Chapter: 6

System of linear Equation

Jacobi method

* Diagonally Dominant:

$$\left. \begin{aligned} 2x - 3y + 5z &= 7 \\ 4x + 7y - 2z &= 5 \\ 8x - y - 3z &= 10 \end{aligned} \right\}$$

Is this system diagonally dominant?

$$2x - 3y + \boxed{5z} = 7, \quad |5| \geq |2| + |-3| = 5$$

$$4x + \boxed{7y} - 2z = 5, \quad |7| \geq |4| + |-2| = 6$$

$$\boxed{8x} - y - 3z = 10, \quad |8| \geq |-1| + |-3| = 4$$

~~$$8x - y - 3z = 10$$~~

~~$$4x + 7y - 2z = 5$$~~

~~$$2x - 3y + \boxed{5z} = 7$$~~

$$|a_{ii}| \geq \sum_{j=1, i \neq j}^n a_{ij}$$

$$4. (a) \quad x + \boxed{8y} + 3z = 10, \quad |8| \geq |1| + |3| = 4$$

$$3x - 5y + 7z = 4, \quad |7| \not\geq |3| + |-5| = 8$$

$$\boxed{3x} - y - z = 1, \quad |3| \geq |-1| + |-1| = 2$$

This is not diagonally dominant.

(i) + (ii)

$$4x + 3y + \boxed{10z} = 14, \quad |10| \geq |4| + |3| = 7$$

(ii) - (iii)

$$0x - 4y + 8z = 3, \quad |8| \geq |0| + |-4| = 4$$

$$\left. \begin{array}{l} \cancel{3x - y - z = 1} \\ \cancel{x + 8y + 3z = 10} \\ \cancel{4x + 3y + 10z = 14} \end{array} \right\}$$

diagonally dominant.

$$(b) \quad 2x + \boxed{10y} - 7z = 20, \quad |10| \geq |2| + |-7| = 9$$

$$3x - 7y - 5z = 18, \quad |7| \not\geq |3| + |-5| = 8$$

$$\boxed{8x} - 5y - 2z = 12, \quad |8| \geq |5| + |-2| = 7$$

(i) + (ii)

$$5x + 3y + \boxed{-12z} = 38, \quad |12| \geq |5| + |3| = 8$$

$$(c) \quad 5x + 9y + 12z = 9, \quad |12| \not\geq |5| + |9| = 14$$

$$8x - 4y - 11z = 14, \quad |11| \not\geq |8| + |-4| = 12$$

$$-2x + \boxed{5y} + z = 10, \quad |5| \geq |-2| + |1| = 3$$

$$(i) + (ii) \quad \boxed{13x} + 5y + z = 23, \quad |13| \geq |5| + |1| = 6$$

② + ③ $6x + y - 10z = 24, \quad | -10 | \geq | 6 | + | 1 | = 7$

Jacobi method:

4(c) $5x + 9y + 12z = 9, \quad 8x - 4y - 11z = 14, \quad -2x + 5y + z = 0$

a. convert the system into diagonally dominant form. Also write the formula for Jacobi iteration.

b. perform 3 iterations with Jacobi method.

c. write a MATLAB code with backslash operator

Solⁿ: ① $\begin{cases} 13x + 5y + z = 23 \\ -2x + 5y + z = 10 \\ 6x + y - 10z = 24 \end{cases}$ } dominant

$$\begin{aligned} x_{n+1} &= \frac{23 - 5y_n - z_n}{13} = \frac{1}{13} (23 - 5y_n - z_n) \\ y_{n+1} &= \frac{1}{5} (10 + 2x_n - z_n) \\ z_{n+1} &= -\frac{1}{10} (24 - 6x_n - y_n) \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Jacobi iterative formula.}$$

② Let, $x_0 = y_0 = z_0 = 0$ or $(x_0, y_0, z_0) = (0, 0, 0)$

put $n=0$ (1st iteration):

$$x_1 = \frac{1}{13} (23 - 5y_0 - z_0) = \frac{1}{13} (23 - 0 - 0) = \frac{23}{13} = 1.769$$

$$y_1 = \frac{1}{5} (10 + 2x_0 - z_0) = \frac{1}{5} (10 + 0 - 0) = 2$$

$$z_1 = -\frac{1}{10} (24 - 6x_0 - y_0) = -\frac{1}{10} (24 - 0 - 0) = -2.4$$

2nd iteration: ($n=1$)

$$x_2 = \frac{1}{13} (23 - 5y_1 - z_1) = \frac{1}{13} (23 - 5 \times 2 - (-2.4)) = 1.185$$

$$y_2 = \frac{1}{5} (10 + 2x_1 - z_1) = \frac{1}{5} (10 + 2(1.769) - (-2.4)) = 3.188$$

$$z_2 = -\frac{1}{10} (24 - 6x_1 - y_1) = -\frac{1}{10} (24 - 6(1.769) - (2)) = 1.139$$

3rd iteration ($n=2$).

$$x_3 = \frac{1}{13} (23 - 5y_2 - z_2) = \frac{1}{13} (23 - 5(3.188) - (1.139)) = 0.455$$

$$y_3 = \frac{1}{5} (10 + 2x_2 - z_2) = \frac{1}{5} (10 + 2(1.185) - (1.139)) = 2.246$$

$$z_3 = -\frac{1}{10} (24 - 6x_2 - y_2) = -\frac{1}{10} (24 - 6(1.185) - (3.188)) =$$

✓

using Calculator

$$x_{n+1} = \frac{1}{13} (23 - 5B - C)$$

$$y_{n+1} = \frac{1}{5} (10 + 2A - C)$$

$$z_{n+1} = -\frac{1}{10} (24 - 6A - B)$$

$$x_0 = y_0 = z_0 = 0$$

1st iteration

$$x_1 = 1.7692$$

$$y_1 = 2$$

$$z_1 = -2.4$$

2nd iteration

$$x_2 = 1.1846$$

$$y_2 = 3.1877$$

$$z_2 = -1.1385$$

3rd

$$x_3 = 0.6308$$

$$y_3 = 2.7015$$

$$z_3 = -1.3705$$

Gauss-seidel Method

Applications:

	Ice-Cream (x)	Milkshake (y)	Cake (z)	consume
Vit. A	20	15	37	= 125
Vit. B	10	25	10	147
Vit. D	35	15	10	200

- a. Find the quantity of Ice cream, Milkshake & cake.
using Gauss-seidel method. (perform 3 iterations).
- b. Write a MATLAB CODE using backslash operator.

Soln:

$$\begin{cases} 20x + 15y + 37z = 125 \\ 10x + 25y + 10z = 147 \\ 35x + 15y + 10z = 200 \end{cases}$$

$$|37| \geq |20| + |15| = 35$$

$$|25| \geq |10| + |10| = 20$$

$$|35| \geq |15| + |10| = 25$$

$$\begin{cases} 35x + 15y + 10z = 200 \\ 10x + 25y + 10z = 147 \\ 20x + 15y + 37z = 125 \end{cases}$$

$$x_{n+1} = \frac{1}{35} (200 - 15y_n - 10z_n)$$

$$y_{n+1} = \frac{1}{25} (147 - 10x_{n+1} - 10z_n)$$

$$x_0 = 0, y_0 = 0, z_0 = 0$$

$$z_{n+1} = \frac{1}{37} (125 - 20x_{n+1} - 15y_{n+1})$$

Iteration 1: (n=0)

$$x_1 = \frac{1}{35} (200 - 15y_0 - 10z_0) = 5.7143$$

$$y_1 = \frac{1}{25} (147 - 10x_1 - 10z_0) = 3.5943$$

$$z_1 = \frac{1}{37} (125 - 20x_1 - 15y_1) = -1.1676$$

iteration 2: (n=1)	}	<u>iteration 3: (n=2)</u>
$x_2 = 4.5075$		$x_3 = 4.0241$
$y_2 = 4.5440$		$y_3 = 4.6305$
$z_2 = -0.9003$		$z_3 = -0.674$

Ans

⑥ MATLAB code:

$$A = \begin{bmatrix} 20 & 15 & 37 \\ 10 & 25 & 10 \\ 35 & 15 & 10 \end{bmatrix}, x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 125 \\ 147 \\ 200 \end{bmatrix}$$

$$Ax = B$$

→ Backslash method

$$\rightarrow A = [20, 15, 37; 10, 25, 10; 35, 15, 10];$$

$$B = [125; 147; 200];$$

$$sol = A \setminus B$$

13.

$$\left. \begin{aligned} \boxed{5x} + 2y + z &= 7 \\ 2x - 4y + 3z &= 6 \\ 3x + 5y + 7z &= 6 \end{aligned} \right\}$$

$$|5| \geq |2| + |1| = 3$$

$$|-4| \not\geq |2| + |3| = 5$$

$$|7| \not\geq |3| + |5| = 8$$

$$x_0 = 1, y_0 = -0.5, z_0 = 0.3$$

$$\textcircled{a} \rightarrow 5x + y + \boxed{10z} = 12, \quad |10| \geq |5| + |1| = 6 \quad \textcircled{ii} + \textcircled{iii}$$

$\textcircled{i} - \textcircled{ii}$

$$3x + \boxed{6y} - 2z = 1, \quad |6| \geq |3| + |-2| = 5$$

Gauss Seidel.

$$\boxed{5x} + 2y + z = 7 \Rightarrow x_{n+1} = \frac{1}{5} (7 - 2y_n - z_n)$$

$$3x + \boxed{6y} - 2z = 1 \Rightarrow y_{n+1} = \frac{1}{6} (1 - 3x_{n+1} + 2z_n)$$

$$5x + y + \boxed{10z} = 12 \Rightarrow z_{n+1} = \frac{1}{10} (12 - 5x_{n+1} - y_{n+1})$$

Jacobi:

$$x_{n+1} = \frac{1}{5} (7 - 2y_n - z_n)$$

$$y_{n+1} = \frac{1}{6} (1 - 3x_n + 2z_n)$$

$$z_{n+1} = \frac{1}{10} (12 - 5x_n - y_n)$$

Jacobi :

iteration 1 ($n=0$) ($x_0 = 1, y_0 = -0.5, z_0 = 0.3$)

$$x_1 = \frac{1}{5} (7 - 2y_0 - z_0) = 1.54$$

$$y_1 = \frac{1}{6} (1 - 3x_0 + 2z_0) = -0.233$$

$$z_1 = \frac{1}{10} (12 - 5x_0 - y_0) = 0.75$$

iteration 2: ($n=1$)

$$x_2 = 1.343$$

$$y_2 = -0.353$$

$$z_2 = 0.453$$

Ans

Gauss-Seidel method:

iteration (1) ($n=0$) $x_0 = 1, y_0 = -0.5, z_0 = 0.3$

$$x_1 = 1.54$$

$$y_1 = -0.503$$

$$z_1 = 0.480$$

iteration 2 ($n=1$)

$$x_2 = 1.505$$

$$y_2 = -0.426$$

$$z_2 = 0.490$$

Ans

Gaussian Elimination with partial pivoting

Solve:

$$\begin{cases} 10x - 2y + 5z = 20 \\ 3x + 4y - 11z = 15 \\ 12x - 8y + 4z = 30 \end{cases}$$

$$\begin{cases} 10x - 3y \\ 10x - 5y \end{cases}$$

Operations	x	y	z	b	Eq no.
	12	-8	4	30	E1
	10	-2	5	20	E2
	3	4	-11	15	E3
1st P.P. → E1/12	1	-0.6667	0.3333	2.5	E4
E2/10	1	-0.2	0.5	2	E5
E3/3	1	1.3333	-3.6667	5	E6
E5-E4	0	0.4667	0.1667	-0.5	E7
E6-E4	0	2	-4	2.5	E8
2nd P.P. → E8/2	0	1	-2	1.25	E9
E7/0.4667	0	1	0.3572	-1.0713	E10
E10-E9	0	0	2.3572	-2.3213	E11
3rd P.P. → E11/-1.3095	0	0	1	-0.9848	E12

3rd P.P.: $(z) = -0.9848$ ✓

2nd P.P.: $y - 2z = 1.25 \Rightarrow y = 1.25 + 2(-0.9848) = -0.7196$

$$1^{st} \text{ p.p.: } x - 0.6667y + 0.333z = 2.5$$

$$x = 2.3495$$

$$\text{--- } 0 \text{ ---}$$

verify your results using substitution method.

$$\begin{aligned} \textcircled{E1} \Rightarrow & 10 \times (0.7727) - 2 \times (-1.7045) + (5 \times 1.7727) \\ & = 19.999 \approx 20 \end{aligned}$$

Chapter 7

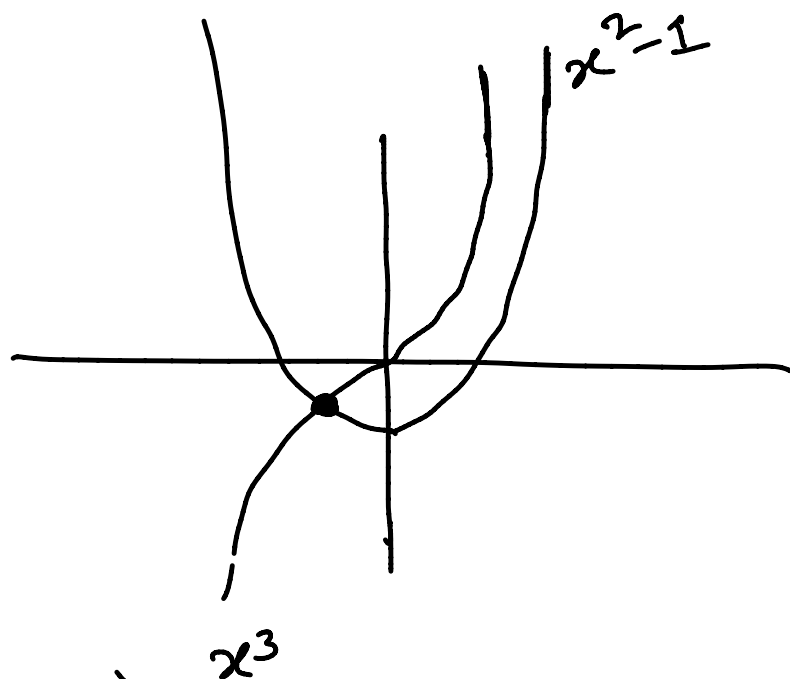
→ Graphical method + Bisection method.

$$\left. \begin{aligned} x^4 - x^2 &= 0 \\ x^2(x^2 - 1) &= 0 \\ x &= 0, x = \pm 1, x = 0 \end{aligned} \right\}$$

$$x^3 - x^2 + 1 = 0$$

$$\Rightarrow x^3 = x^2 - 1$$

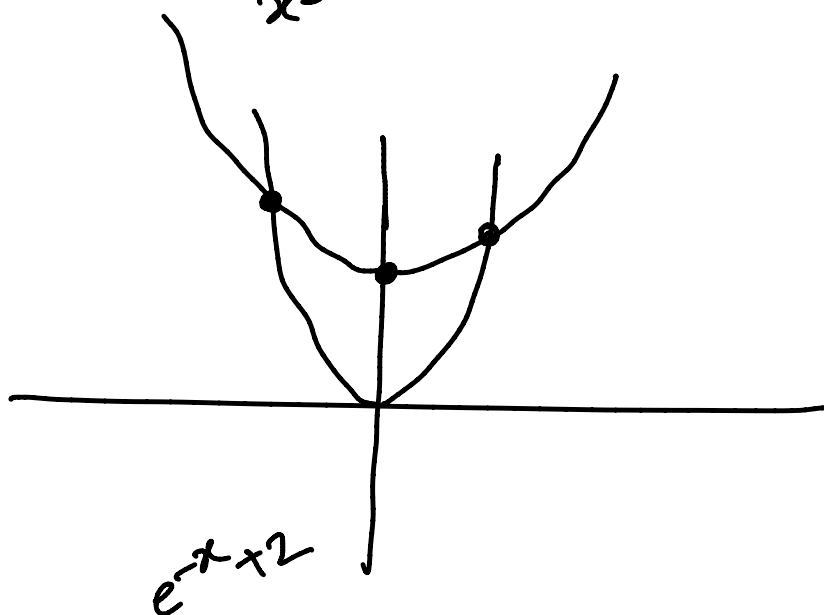
$$\left. \begin{aligned} \text{real root} &= 1 \\ \text{complex} &= 2 \end{aligned} \right\}$$



$$\# \quad x^4 - x^2 - 1 = 0$$

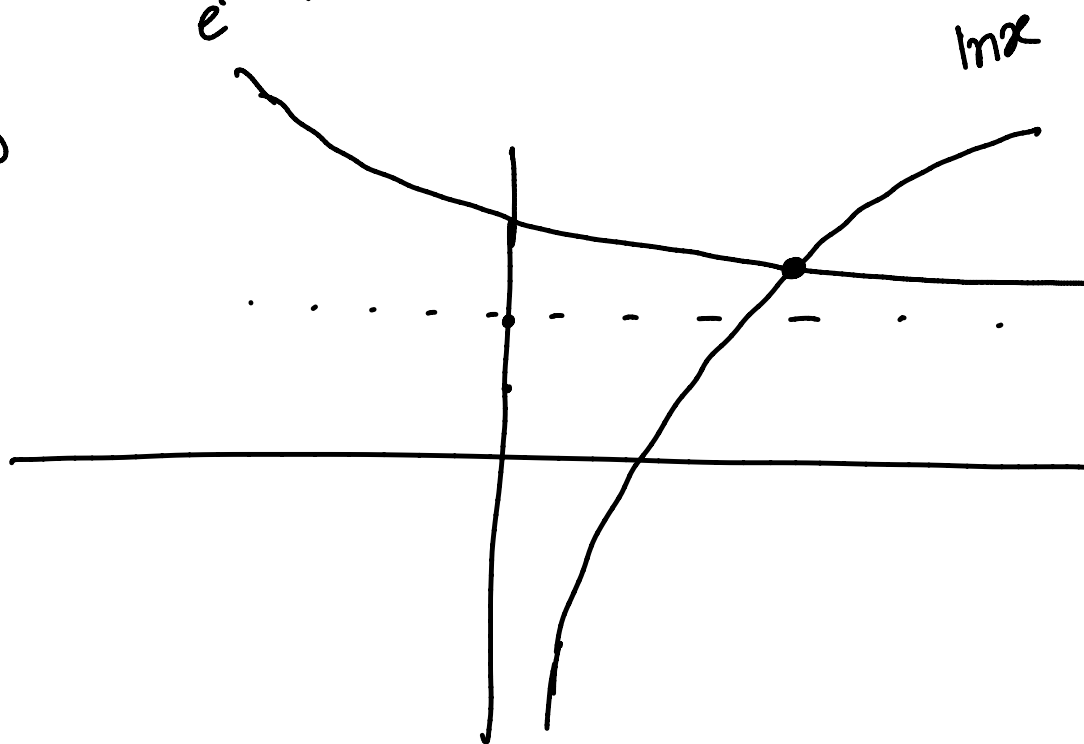
$$\Rightarrow x^4 = x^2 + 1$$

$$\left. \begin{aligned} \text{real} &= 2 \\ \text{complex} &= 2 \end{aligned} \right\}$$



$$\# \quad e^{-x} - \ln x + 2 = 0$$

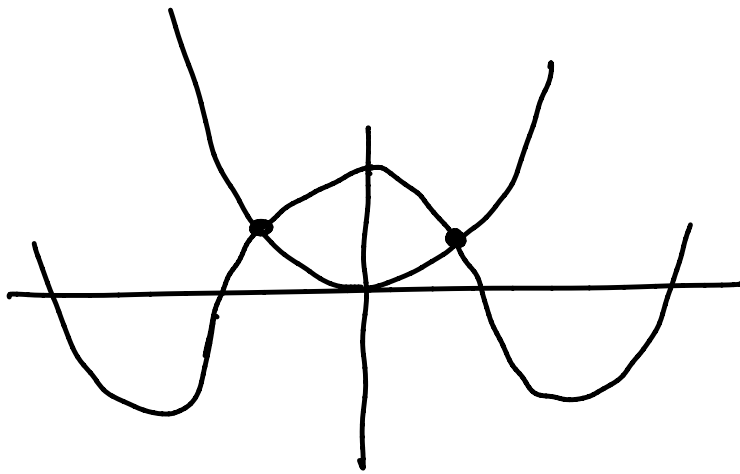
$$\Rightarrow e^{-x} + 2 = \ln x$$



$$\# \quad x^2 - \cos 2x = 0$$

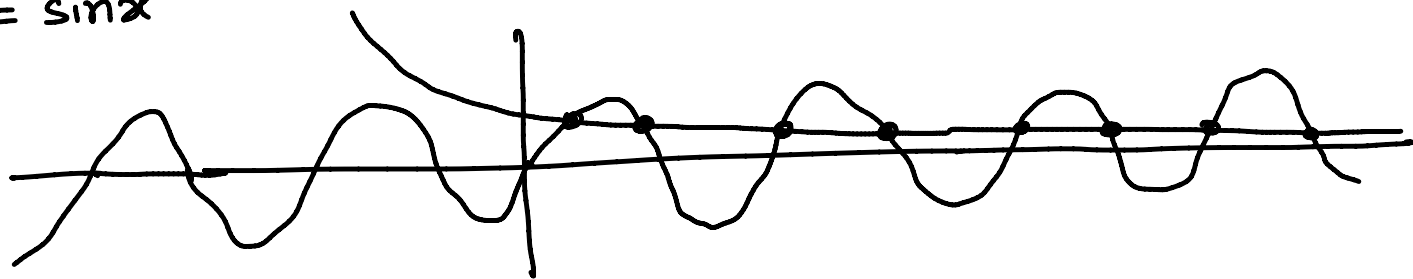
real = 2

$$\Rightarrow x^2 = \cos 2x$$



$$\# \quad e^{-x} - \sin x = 0$$

$$\Rightarrow e^{-x} = \sin x$$

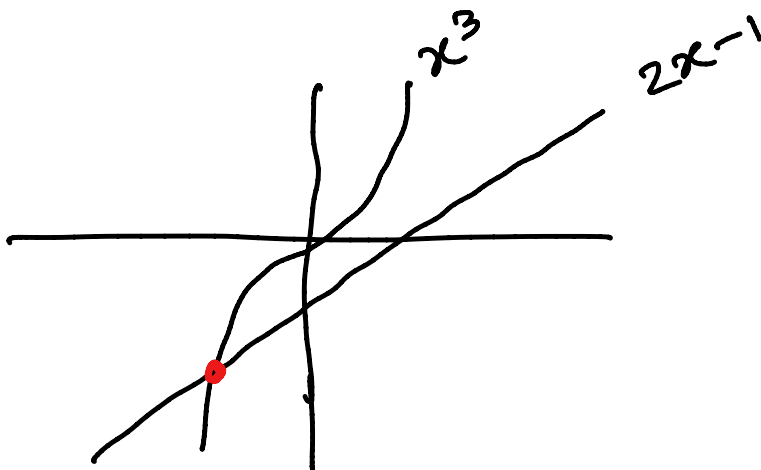


many solutions.

$$\textcircled{\#} \quad x^3 - 2x + 1 = 0$$

$$x^3 = 2x - 1$$

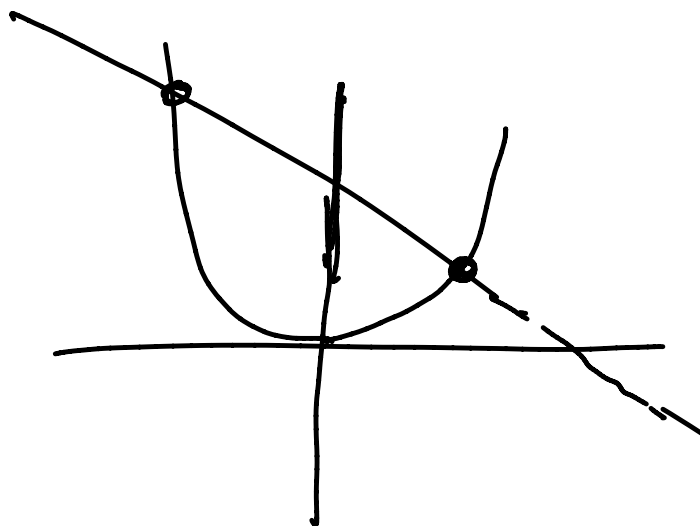
real = 1
complex = 2



$$\# \quad x^2 + 3x - 2 = 0$$

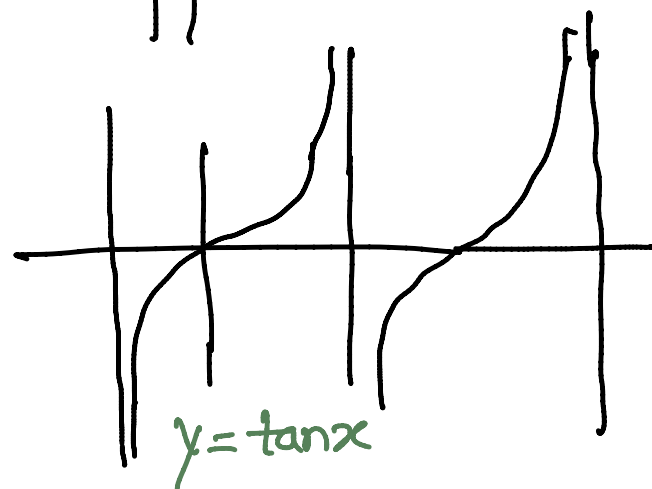
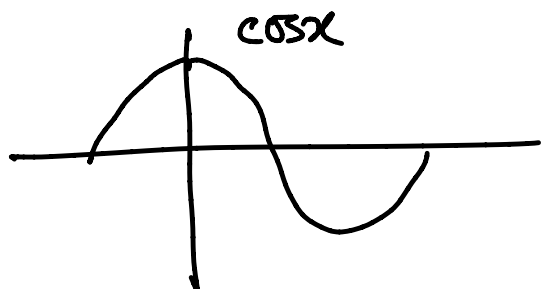
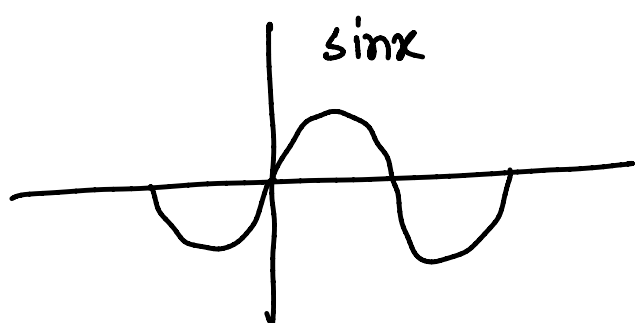
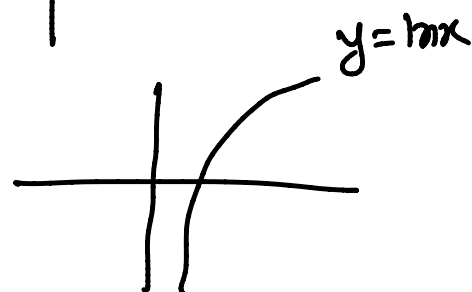
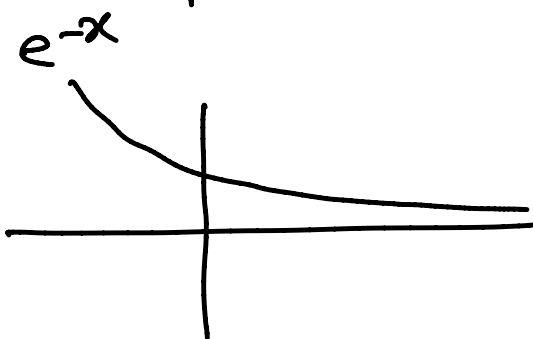
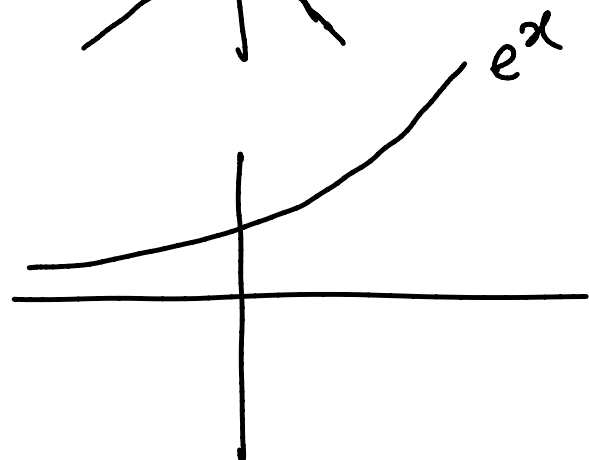
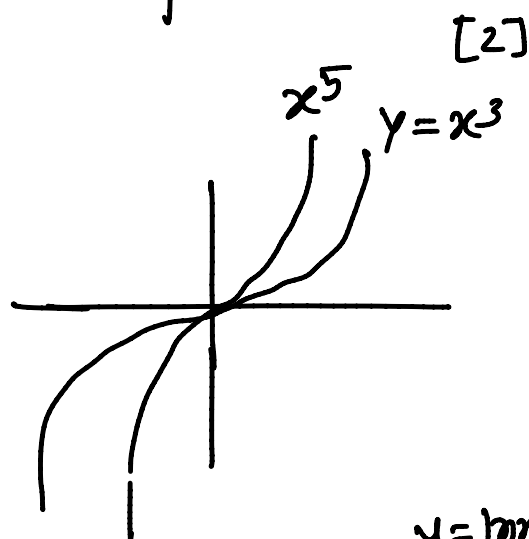
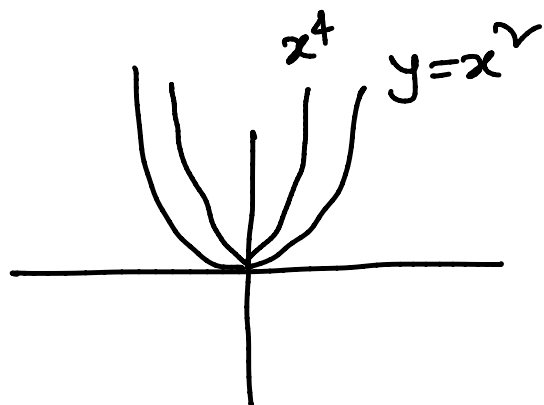
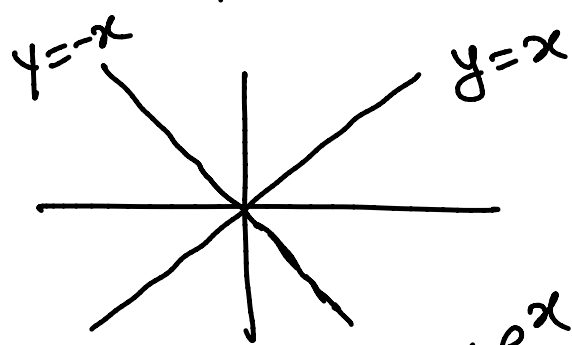
$$\Rightarrow x^2 = 2 - 3x$$

real = 2
complex = 0



locate the root. Hence find the number of real &

complex root(s).

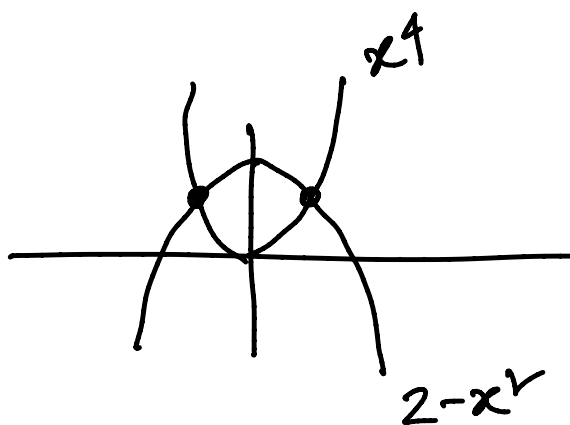


$$x^4 + x^2 - 2 = 0$$

$$\Rightarrow x^4 = 2 - x^2$$

$$\text{real} = 2$$

$$\text{complex} = 2$$



$$x^4 - \cos 2x = 2$$

$$f(x) = x^4 - \cos 2x - 2$$

MATLAB codes: "roots"

$$\rightarrow x^4 - 4x^2 + 2x - 3 = 0$$

$$\rightarrow p = [1, 0, -4, 2, -3];$$

$$\text{roots}(p)$$

$$x^9 + 0x^8 + 0x^7 + 0x^6 - 7x^5$$

$$\rightarrow x^9 - 7x^5 - 2x + 1 = 0$$

$$p = [1, 0, 0, 0, -7, 0, 0, 0, -2, 1];$$

$$\text{roots}(p)$$

Bisection method

→ If $f(a) * f(b) < 0$ then $\exists$ a root on (a, b)
 $\downarrow$
 there exist

Q. Given that, $x^2 - \cos 2x = 0$, $\begin{matrix} [0, 1] \\ \uparrow_a \uparrow_b \end{matrix}$

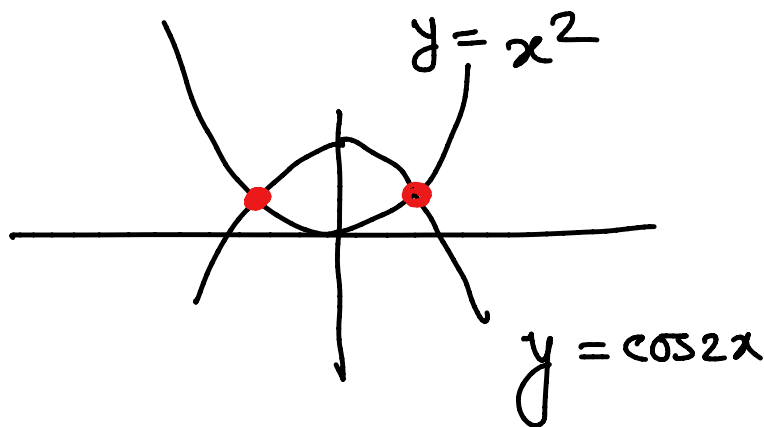
(a) locate the root. & show that there exist a root on the given interval [3]

(b) perform Bisection method 3 times. [3]

(c) write a MATLAB code using "fzero" [2]

Soln:

(a) $x^2 - \cos 2x = 0$
 $x^2 = \cos 2x$



$f(x) = x^2 - \cos 2x$,

$\check{a} = 0, f(\check{a}) = f(0) = -1 < 0$

$b = 1, f(b) = f(1) = 1.4161 > 0$

$\exists$ a root on $(0, 1)$

(b)

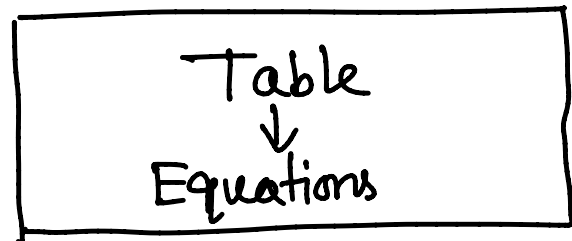
	a	b	$c = \frac{a+b}{2}$	$f(c)$	update
1	0	1	0.5	-0.2903	$a = c$
2	0.5	1	0.75	0.4918	$b = c$
3	0.5	0.75	0.625	0.075	$b = c$
	0.5	0.625			

③ $f = @ (x) x^2 - \cos(2*x);$
 $sol = fzero(f, [0, 1])$

Quiz-01

1. Graphical + Bisection $\rightarrow 8$

2.



① Diagonally dominant.

② Jacobi

③ Gauss - seidel

④ Gaussian Elimination

11th August
Tuesday

Bisection Method

$$\# \sin^{-1}x - x^2 + 1 = 0$$

$$a = -1 \quad f(a) < 0$$

$$b = -0.5 \quad f(b) > 0$$

$\exists$ a root on $(-1, -0.5)$



a	b	$c = \frac{a+b}{2}$	$f(c)$	update
-1	-0.5	-0.75	-0.4	$a = c$
-0.75	-0.5	-0.625	-0.065	$a = c$
-0.625	-0.5	-0.5625	0.086	$b = c$
-0.625	-0.5625			

$\uparrow$
a

$\uparrow$
b

$$c \approx -0.5625$$

secant method

Derivation:

$$y - y_2 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_2)$$

for which x , $\Rightarrow f(x) = 0 = y$

$$\text{put } y = 0 \Rightarrow$$

$$x = x_2 + \frac{x_2 - x_1}{y_2 - y_1} (0 - y_2)$$

$$x = x_2 - \frac{x_2 - x_1}{y_2 - y_1} y_2$$

$$x_{n+1} = B - \frac{B-A}{f(B)-f(A)} f(B) \rightarrow \text{secant method } +O(h)$$

$$\# \sin^{-1}x - x^2 + 1 = 0, \quad [-1, -0.5]$$

→ a. perform Bisection method three times to find new smaller interval. → $a = -0.625, b = -0.5625$

b. perform secant method three " to find root up to 3 d.p. (using a).

⑥
→ soln: $x_{n+1} = b - \frac{b-a}{f(b)-f(a)} f(b)$

$$f(x) = \sin^{-1}(x) - x^2 + 1$$

$$\therefore x_{n+1} = b - \frac{b-a}{(\sin^{-1}b - b^2 + 1) - (\sin^{-1}a - a^2 + 1)} x$$

$$(\sin^{-1}b - b^2 + 1)$$

it	x_{n+1}
a	-0.625
b	-0.5625
...	...
1	-0.5980
2	-0.5986
3	-0.5986

$$x \approx -0.5986$$

$$x \approx -0.599 \text{ upto 3.d.p.}$$

$e^{-3x} - \ln(2x) + 1 = 0$, $[1, 2]$

solve using secant method.

Soln: $x_{n+1} = b - \frac{b-a}{f(b)-f(a)} f(b)$

$a=1$
 $b=2$, $f(x) = e^{-3x} - \ln(2x) + 1$

$$x_{n+1} = b - \frac{b-a}{(e^{-3b} - \ln(2b) + 1) - (e^{-3a} - \ln(2a) + 1)} \times (e^{-3b} - \ln(2b) + 1)$$

itt	x_{n+1}
a	1
b	2
1	1.4816
2	1.3567
3	1.3819
4	1.3809
5	1.3809

$x \approx 1.3809$ upto 4 d.p.

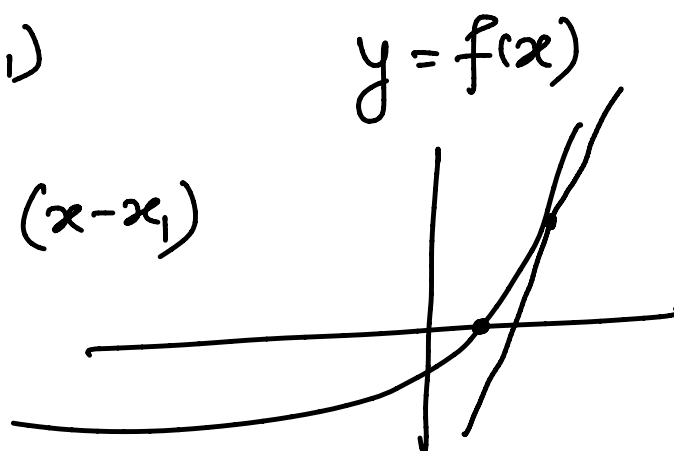
Newton Rapshon method

Derivation: $y - y_1 = m(x - x_1)$

$$\Rightarrow y - y_1 = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} (x - x_1)$$

$y = 0$

$$\Rightarrow -y_1 = f'(x) (x - x_1)$$



$$\Rightarrow x - x_1 = - \frac{y_1}{f'(x)}$$

$$\Rightarrow x = x_1 - \frac{f(x)}{f'(x)}$$

$$\therefore x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

for calculator use.

$$x_{n+1} = a - \frac{f(a)}{f'(a)}$$

$$\# \cosh(x) - x^2 = 0$$

Solve using Newton-Rapshon method.

Soln: $f(x) = \cosh(x) - x^2$

$$f'(x) = \sinh(x) - 2x$$

$$\therefore x_{n+1} = a - \frac{\cosh(a) - a^2}{\sinh(a) - 2a}$$

itt	x_{n+1}
a	1
...	...
1	1.6584
2	1.6207
3	1.6213
4	1.6213

$x \approx 1.6213$ Ans

→ Assume

0.5

2.3326

2.8801

2.6746

2.6028

2.5940

2.5939

$$x \approx \boxed{2.5939}$$

#

$$f(x) = \frac{x^2}{2} - e^{-x^2} + \sin(x) = 0$$

$$f'(x) = \frac{2x}{2} + 2x e^{-x^2} + \cos(x)$$

}

Fixed point iteration (superlinear)



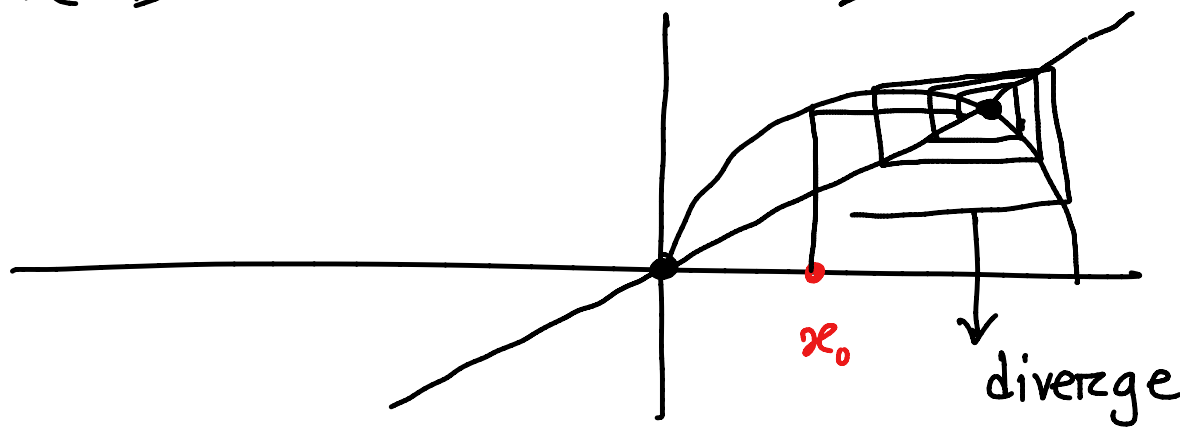
golden ratio

$$\phi \left(\frac{1+\sqrt{5}}{2} \right)$$

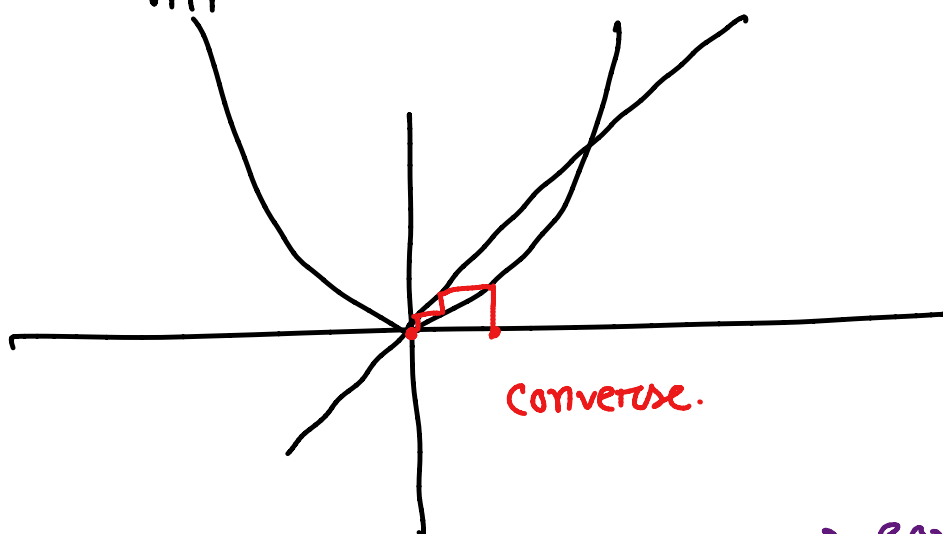
$x_{n+1} = f(x_n)$

$$x_{n+1} = \sin(x_n)$$

Cobweb



$$x_{n+1} = x_n^2$$



$$x_{n+1} = f(x_n)$$

converge if $|f'(x_0)| < 1$
diverge if $|f'(x_0)| > 1$

Example: $x^2 - \cos(2x) = 0$

(i) $x^2 = \cos 2x$

$$x = \sqrt{\cos 2x}$$

$$x_{n+1} = \sqrt{\cos 2x_n}$$

(ii) $\cos 2x = x^2$

$$\therefore 2x = \cos^{-1}(x^2)$$

$$x = \frac{1}{2} \cos^{-1}(x^2)$$

$$x_{n+1} = \frac{1}{2} \cos^{-1}(x_n^2)$$

$$x_0 = 0.7$$

(a) which method will converge near $x_0 = 0.7$

(i) $x_{n+1} = g(x_n)$

$$g(x) = \sqrt{\cos 2x} = (\cos 2x)^{1/2}$$

$$g'(x) = \frac{1}{2} (\cos 2x)^{\frac{1}{2}-1} (-\cancel{2} \sin 2x) = \frac{-\sin 2x}{\sqrt{\cos 2x}}$$

$$|g'(0.7)| = \left| -\frac{\sin(2 \times 0.7)}{\sqrt{\cos(2 \times 0.7)}} \right| = |-2.39| = 2.39 > 1$$

diverge!

$$x_{n+1} = \frac{1}{2} \cos^{-1}(x^2)$$

(ii) $x_{n+1} = g(x_n)$

$$g(x) = \frac{1}{2} \cos^{-1}(x^2)$$

$$g'(x) = \frac{1}{2} \left(\frac{-1}{\sqrt{1-x^4}} \right) \cdot (\cancel{2}x) = \frac{-x}{\sqrt{1-x^4}}$$

$$|g'(0.7)| = |-0.803| = 0.803 < 1$$

converge!

(b) perform 4 iterations using fixed points.

it	x_{n+1}
0	0.7
1	0.5294
2	0.6434
3	0.5720
4	0.6187

$$x_{n+1} = \frac{1}{2} \cos^{-1}(x_n^2)$$

0.5890

0.6083

0.5959

0.6039

0.5987

0.6021

0.5999

0.6013

0.6004

0.6010

0.6006

0.6009

$x \approx 0.601$ upto 3 d.p.

MATLAB:

```
clc
```

```
clear
```

```
x(1)=0.7
```

```
for i = 1:15
```

```
    x(i+1) = (1/2) * acos(x(i)^2);
```

```
end
```

```
while r==1
```

```
do
```

```
    x(i+1) = . . . . .
```

```
    if abs(x(i+1) - x(i)) < 0.00001
```

```
        r=1
```

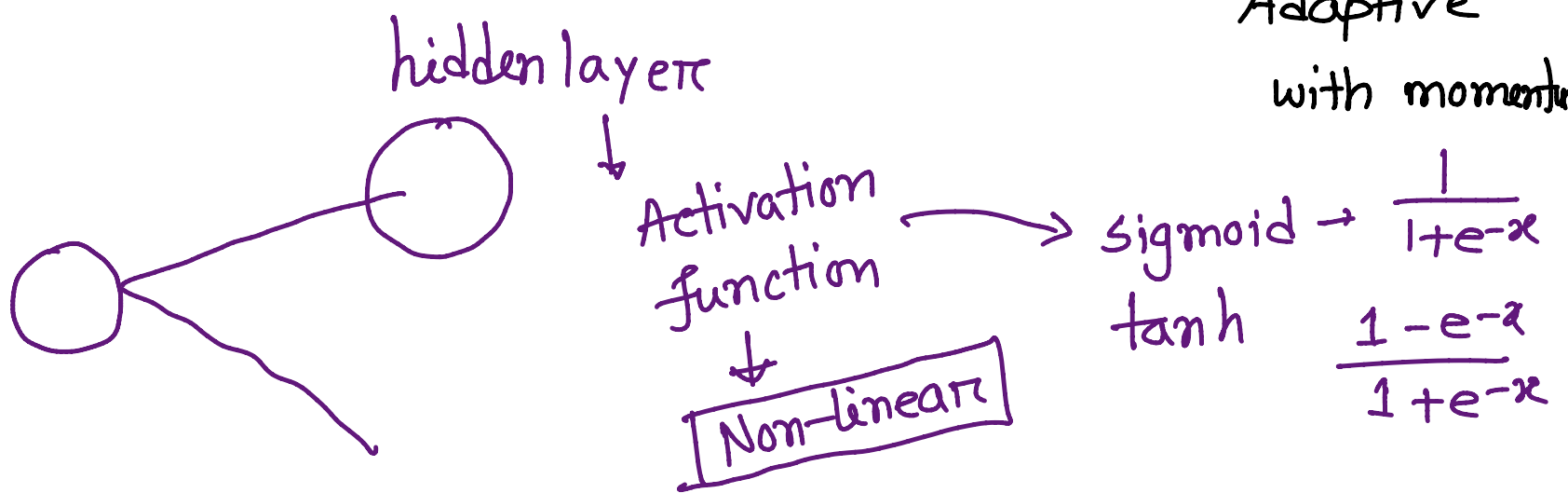
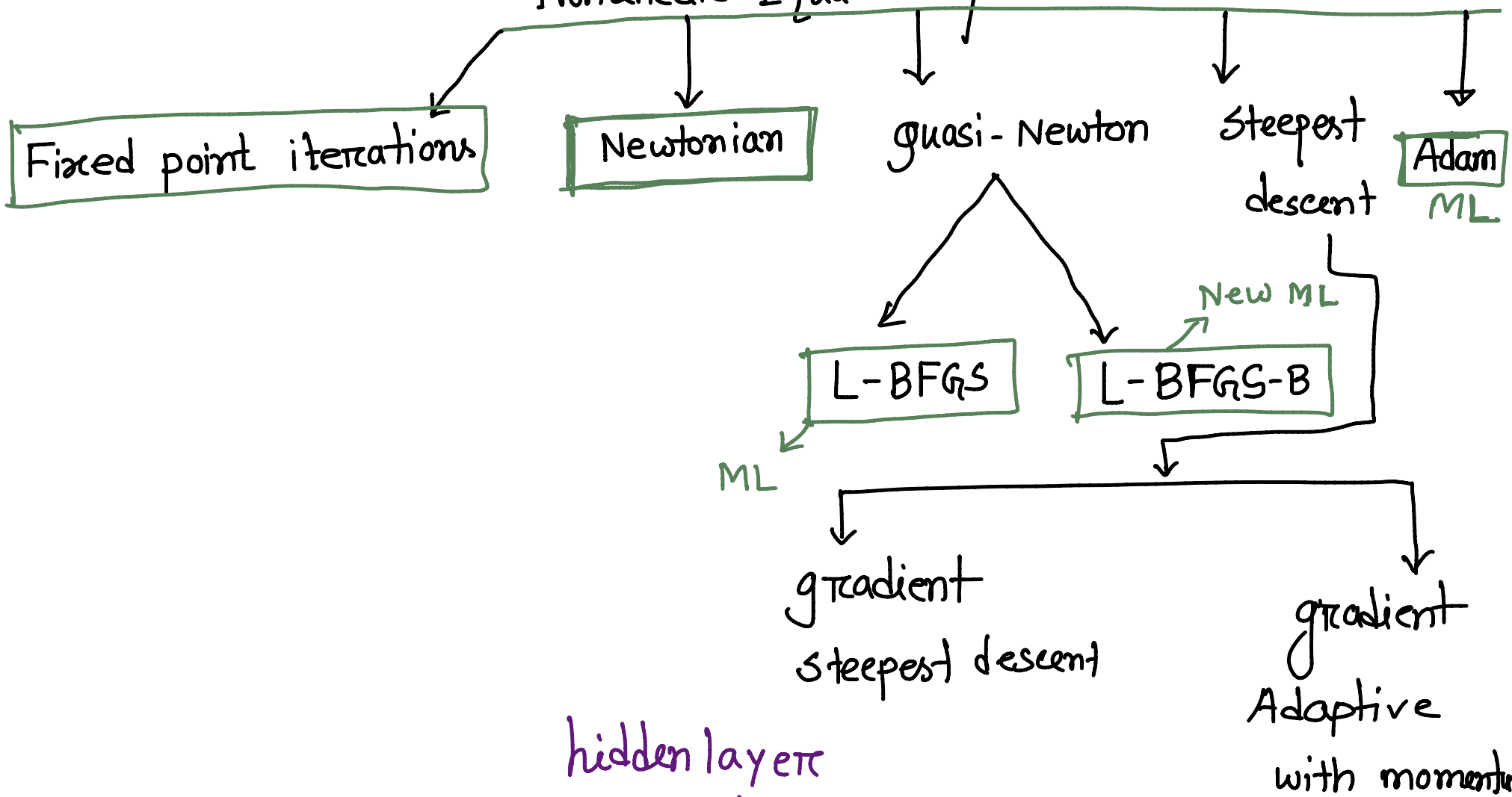
```
    else
```

```
        r=0
```

```
    end
```

Chapter : 8

Nonlinear Equations of two or more variables



Fixed point iteration:

Given that:

$$\left. \begin{aligned} x^2 + y^2 - 10x + 5y - 2 &= 0 \\ xy^2 - 2x + 15y + 5 &= 0 \end{aligned} \right\} \times \rightarrow f_{\text{solve}}$$

To solve the above problem with Fixed point iteration, consider an iterative formula:

$$\left\{ \begin{aligned} x_{n+1} &= \frac{1}{10} (x_n^2 + y_n^2 + 5y_n - 2) \\ y_{n+1} &= -\frac{1}{15} (x_n y_n^2 - 2x_n + 5) \end{aligned} \right\} \rightarrow \text{iterative}$$

- a. Show that, the above iterative method will converge near $(1, 0.5)$. *otherwise state the reason.*
- b. perform 3 iteration with Gauss - Seidel technique.
- c. Write a MATLAB code with "fsolve".
- d. " " " " " iterative technique.

Theorem: If $x_{n+1} = f(x_n, y_n)$ then fixed point
 $y_{n+1} = g(x_n, y_n)$

method converge iff $\left| \frac{\partial f}{\partial x} \right| + \left| \frac{\partial f}{\partial y} \right| < 1$ &
 $\left| \frac{\partial g}{\partial x} \right| + \left| \frac{\partial g}{\partial y} \right| < 1$

near (x_0, y_0) .

soln: (a) $x_{n+1} = \frac{1}{10} (x_n^2 + y_n^2 + 5y_n - 2) = f(x_n, y_n)$
 $y_{n+1} = -\frac{1}{15} (x_n y_n^2 - 2x_n + 5) = g(x_n, y_n)$

$$\begin{aligned} f(x, y) &= \frac{1}{10} (x^2 + y^2 + 5y - 2) & g(x, y) &= -\frac{1}{15} (x y^2 - 2x + 5) \\ \frac{\partial f}{\partial x} &= \frac{1}{10} (2x + 0) = \frac{x}{5} & \frac{\partial g}{\partial x} &= -\frac{1}{15} (y^2 - 2) = \frac{2 - y^2}{15} \\ \frac{\partial f}{\partial y} &= \frac{1}{10} (0 + 2y + 5) = \frac{2y + 5}{10} & \frac{\partial g}{\partial y} &= -\frac{1}{15} (2xy + 0 + 0) \\ & & &= -\frac{2xy}{15} \end{aligned}$$

$$(x_0, y_0) = (1, 0.5)$$

$$\left| \frac{\partial f}{\partial x} \right| + \left| \frac{\partial f}{\partial y} \right| = \left| \frac{1}{5} \right| + \left| \frac{2 \times 0.5 + 5}{10} \right| = |0.2| + |0.6| = 0.8 < 1$$

$$\left| \frac{\partial g}{\partial x} \right| + \left| \frac{\partial g}{\partial y} \right| = \left| \frac{2 - (0.5)^2}{15} \right| + \left| -\frac{2 \times 1 \times 0.5}{15} \right| = |0.1167| + |-0.067|$$

$$= 0.1167 + 0.067 < 1$$

$\therefore$ It will converge.

b.

	$x_{n+1} = \frac{1}{10} (x_n^2 + y_n^2 + 5y_n - 2)$ ✓	$y_{n+1} = -\frac{1}{15} (x_{n+1} y_n^2 - 2x_{n+1} + 5)$ ✓
0	$x_0 = 1$	$y_0 = 0.5$
1	0.175 ✓	-0.3129 ✓
2	-0.3436 ✓	-0.3769 ✓
3	-0.3624	-0.3782

$x \approx -0.3624$
 $y \approx -0.3782$

} After 3rd iteration

4 -0.3616 -0.3781

5 -0.3617 -0.3781

$x \approx -0.3617$

$y \approx -0.3781$

correct up to
3 d.p.

c. fsolve:

$$\begin{cases} x^2 + y^2 - 10x + 5y - 2 = 0 \\ xy^2 - 2x + 15y + 5 = 0 \end{cases}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$X(1) = x$$

$$X(2) = y$$

$$\left. \begin{array}{l} x_0 = 1 \\ y_0 = 0.5 \end{array} \right\}$$

$$X(1)^2 + X(2)^2 - 10 * X(1) + 5 * X(2) - 2$$

→ Matlab code: fsolve(fun, x0)

$$f = @(x) \begin{bmatrix} x(1)^2 + x(2)^2 - 10 * x(1) + 5 * x(2) - 2; \\ x(1) * x(2)^2 - 2 * x(1) + 15 * x(2) + 5 \end{bmatrix}$$

$$sol = fsolve \left(f, \begin{bmatrix} \underline{1} \\ \underline{0.5} \end{bmatrix} \right)$$

↑
x

↑
y

Chapter 7 MATLAB codes:

(i) roots(p): only for polynomial

$$x^5 - 2x^4 + 5x - 7 = 0$$

$$\begin{cases} p = [1, 0, 0, -2, 5, -7] \\ \text{roots}(p) \end{cases}$$

$$x^9 - 1 = 0$$

$$\begin{cases} p = [1, 0, 0, 0, 0, 0, 0, 0, 0, -1] \\ \text{roots}(p) \end{cases}$$

iii) fsolve (any type):

$$\# x^2 - \cos 2x = 0.$$

$$\begin{cases} f = @(x) x^2 - \cos(2*x) \\ \text{fsolve}(f, \underline{0.5}) \end{cases}$$

iv) iterative method:

$$x_{n+1} = \frac{1}{2} \cos^{-1}(x_n) \quad x_0 = 0.7$$

↗ $\text{acos}(x)$

$$x(1) = 0.7$$

$$\text{for } i = 1: 10$$

$$x(i+1) = (1/2) * \text{acos}(x(i));$$

end

ii) fzero (any function where roots exist on $[a, b]$)

$$x^2 - \cos(2x) = 0, [0, 1]$$

$$\begin{cases} f = @(x) x^2 - \cos(2*x) \\ \text{fzero}(f, [0, 1]) \end{cases}$$

$$e^{-3x} + 4x \ln(x) + \tan x = 2$$
$$[0, 5]$$

$$\begin{cases} f = @(x) \exp(-3*x) + 4*x*\log(x) + \tan(x) - 2 \\ \text{fzero}(f, [0, 5]) \end{cases}$$

ch.7 (review)

Secant Method: Solve $x^4 - 2x + 4 = 0$ using secant method on $[1, 2]$.

Solⁿ: Given that, $f(x) = 0$
 $x^4 - 2x + 4 = 0$
 $f(x) = x^4 - 2x + 4$

$$a = 1, b = 2$$

$$x_{n+1} = b - \frac{b - a}{f(b) - f(a)} \times f(b) \Rightarrow$$

$$x_{n+1} = b - \frac{b - a}{(b^4 - 2b + 4) - (a^4 - 2a + 4)} \times (b^4 - 2b + 4) \quad \text{From (i)}$$

1st iteration ($n=0$) : $a = 1$
 $b = 2$

$$x_1 = \boxed{} \quad \text{— From (i)}$$

2nd iteration ($n=1$) : $a = 2, b = x_1$

From (i)

$$x_2 =$$

3rd iteration ($n=2$) : $a = x_1, b = x_2$

From (i)

$$x_3 =$$

Newton's method:

$$e^{-3x} - \ln(2x) - 1 = 0, \quad \boxed{x_0 = 1}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Soln: $f(x) = e^{-3x} - \ln(2x) - 1$

$$f'(x) = -3e^{-3x} - \frac{1}{2x} \cdot 2 = -3e^{-3x} - \frac{1}{x}$$

1st iteration:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_0 = 1$$

$$f(x_0) = e^{-3} - \ln(2) - 1 = -1.6434$$

$$f'(x_0) = -3e^{-3} - 1 = -1.1494$$

$$\therefore x_1 = 1 - \frac{-1.6434}{-1.1494}$$

$$= -0.4297$$

2nd iteration:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_1 = -0.4297$$

$$f(x_1) = e^{\pi\pi\pi}$$

$$f'(x_1) =$$

$$\therefore x_2 =$$

$$x_0 = 0.5$$

1st iteration:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{-0.7768}{-2.6694}$$

$$x_0 = 0.5$$

$$= 0.2089$$

$$f(x_0) = -0.7768$$

$$f'(x_0) = -2.6694$$

2nd iteration:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_1 = 0.2089$$

$$f(x_1) = 0.4071$$

$$f'(x_1) = -6.3900$$

$$x_2 = 0.2089 - \frac{0.4071}{-6.3900}$$

$$= 0.2726$$

$$\# \quad x_{n+1} = a - \frac{e^{-3a} - \ln(2a) - 1}{-3e^{-3a} - \frac{1}{a}} \quad (\text{calculator})$$

itt	x_{n+1}
0	0.5
1	0.2089
2	0.2726
3	0.2822
4	0.2824
5	0.2824

1st iteration: (n=0)

$$x_1 = \frac{1}{10} (x_0^2 + y_0^2 - 5y_0 - 2) = \frac{1}{10} (1 + 0.5^2 - 5 \times 0.5 - 2) \\ = \boxed{0.175}$$

$$y_1 = \frac{-1}{15} (x_1 y_0^2 - 2x_1 + 5)$$

$$= -\frac{1}{15} (0.175 \times 0.5^2 - 2 \times 0.175 + 5)$$

2nd iter .

.

,

,

Ch. 8

Newtonian method for solving two or more systems:

$$\rightarrow x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad F(x) = \begin{bmatrix} f(x,y) \\ g(x,y) \end{bmatrix}$$

$$\rightarrow J(x) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} \quad J \rightarrow \text{Jacobian Matrix}$$

$$\rightarrow x_{n+1} = x_n - [J(x_n)]^{-1} F(x_n) \quad x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$

for more than two variables the above idea can be extended.

Example 1: Consider $x^v + y^v = 9$
 $y^v = 4x$

$$(x_0, y_0) = (2, 2.5)$$

a. perform Newton's method one time to find next iteration value.

b. Write a MATLAB code using "fsolve(fun, x0)"

$$\text{Soln: } \left. \begin{array}{l} x^v + y^v - 9 = 0 \\ y^v - 4x = 0 \end{array} \right\} \therefore \begin{array}{l} f(x,y) = x^v + y^v - 9 \\ g(x,y) = y^v - 4x \end{array}$$

$$\text{Let, } x = \begin{bmatrix} x \\ y \end{bmatrix} \quad F(x) = \begin{bmatrix} f(x,y) \\ g(x,y) \end{bmatrix} = \begin{bmatrix} x^v + y^v - 9 \\ y^v - 4x \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial g}{\partial x} = -4, \quad \frac{\partial g}{\partial y} = 2y$$

$$J(x) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ -4 & 2y \end{bmatrix}$$

1st iteration (n=0)

$$x_1 = x_0 - [J(x_0)]^{-1} F(x_0)$$

$$= \begin{bmatrix} 2 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 4 & 5 \\ -4 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 4 + 6.25 - 9 \\ 6.25 - 4 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 4 & 5 \\ -4 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 1.25 \\ -1.75 \end{bmatrix}$$

$$= A - B^{-1}C$$

$$x_1 = \begin{bmatrix} 1.625 \\ 2.55 \end{bmatrix} \quad \underline{\text{Ans}}$$

$$x_2 = x_1 - [J(x_1)]^{-1} F(x_1)$$

⋮

MATLAB:

$$x = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x(1) = x$$

$$x(2) = y$$

$$x^2 + y^2 - 9 = 0$$

$$y^2 - 4x = 0$$

$$\begin{cases} f = @(x) [x(1)^2 + x(2)^2 - 9; x(2)^2 - 4 * x(1)] \\ \text{sol} = \text{fsolve}(f, [2; 2.5]) \end{cases}$$

problem 2: Given that, $x^{\sim} + y^{\sim} - 10x + 5y - 5 = 0$
 $xy^{\sim} - x^{\sim}y + 5x - 2y + 2 = 0$

$$(x_0, y_0) = (1.5, 2)$$

a. perform Newton method once to find next iterative value.

b. MATLAB code.

Soln: $f(x, y) = x^{\sim} + y^{\sim} - 10x + 5y - 5$

$$g(x, y) = xy^{\sim} - x^{\sim}y + 5x - 2y + 2$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} \quad F = \begin{bmatrix} x^{\sim} + y^{\sim} - 10x + 5y - 5 \\ xy^{\sim} - x^{\sim}y + 5x - 2y + 2 \end{bmatrix}$$

$$\left. \begin{aligned} \frac{\partial f}{\partial x} &= 2x + 0 - 10 = 2x - 10 \\ \frac{\partial f}{\partial y} &= 0 + 2y + 0 + 5 = 2y + 5 \end{aligned} \right\}$$

$$\left. \begin{aligned} \frac{\partial g}{\partial x} &= y^2(1) - y(2x) + 5 = y^{\sim} - 2xy + 5 \\ \frac{\partial g}{\partial y} &= x(2y) - x^2 \cdot 1 + 0 - 2 = 2xy - x^{\sim} - 2 \end{aligned} \right\}$$

$$J(x) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x - 10 & 2y + 5 \\ y^{\sim} - 2xy + 5 & 2xy - x^{\sim} - 2 \end{bmatrix}$$

1st iteration ($n=0$):

$$x_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix}$$

$$x_1 = x_0 - [J(x_0)]^{-1} F(x_0)$$

$$= \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} - \begin{bmatrix} -7 \\ 3 \end{bmatrix} \begin{bmatrix} 9 & -3.75 \\ 1.75 & 7 \end{bmatrix}^{-1} \begin{bmatrix} -3.75 \\ 7 \end{bmatrix}$$

$$= A - B^{-1}C$$

$$x_1 = \begin{bmatrix} -0.2723 \\ 1.0382 \end{bmatrix}$$

Ans

MATLAB:

$$f = @(x) \begin{bmatrix} x(1)^2 + x(2)^2 - 10 * x(1) + 5 * x(2) - 5; \\ x(1) * x(2)^2 - x(1)^2 * x(2) + 5 * x(1) - 2 * x(2) + 2 \end{bmatrix};$$

$$sol = fsolve(f, [1.5; 2])$$

Ans

Data
 $\Downarrow$
 Function

$\Downarrow$
 missing value

Interpolation

Newtonian
 Dividend
 Method

Lagrangian
 Interpolation

#

α	1°	3°	7°	10°
C_L	0.562	0.713	0.945	1.112



Airfoil

α = angle of attack

C_L = Lift Force

a. use Lagrangian Interpolation to find

two points Lift force (C_L) at $\alpha = 4^\circ$.

b. use Linear interpolation to find α at $C_L = 0.82$.

c. use Lagrangian " " " " α at $C_L = 1$ (3 points)

d. Write a MATLAB code using "polyfit" & "polyval" to compute C_L at $\alpha = 1.5^\circ, 2^\circ, 8^\circ$.

Soln:

$x(\alpha)$	1	3	7	10
$y(C_L)$	0.562	0.713	0.945	1.112

$$y = \frac{(x-3)(x-7)(x-10)}{(1-3)(1-7)(1-10)} (0.562) + \frac{(x-1)(x-7)(x-10)}{(3-1)(3-7)(3-10)} (0.713) \\ + \frac{(x-1)(x-3)(x-10)}{(7-1)(7-3)(7-10)} (0.945) + \frac{(x-1)(x-3)(x-7)}{(10-1)(10-3)(10-7)} (1.112)$$

Put

$$\alpha = x = 4$$

$$y = \frac{(4-3)(4-7)(4-10)}{(-2)(-6)(-9)} (0.562) + \frac{(4-1)(4-7)(4-10)}{(2)(-4)(-7)} (0.713) \\ + \frac{(4-1)(4-3)(4-10)}{(6)(4)(-3)} (0.945) + \frac{(4-1)(4-3)(4-7)}{9 \cdot 7 \cdot 3} (1.112)$$

$$\approx \boxed{-0.0937} + \boxed{0.6875} + \boxed{0.4725} + \boxed{-0.0529}$$

$$= \boxed{1.0134}$$

Ans

(b)

α	3	7
C_L	0.713	0.915

$$C_L = 0.82$$

$$\alpha = ?$$

$$\Downarrow$$

(y)	C_L	0.713	0.915
(x)	α	3	7

$$x = \frac{(y - 0.915)}{(0.713 - 0.915)} (3) + \frac{(y - 0.713)}{(0.915 - 0.713)} (7)$$

$$x = \frac{(0.82 - 0.915)}{(0.713 - 0.915)} (3) + \frac{(0.82 - 0.713)}{(0.915 - 0.713)} (7)$$

$$= \boxed{}$$

Ans

d c.

(y)	c_L	0.713	0.915	1.112
(x)	α	3	7	10

$$x = \frac{(y-0.915)(y-1.112)}{(0.713-0.915)(0.713-1.112)} (3) + \frac{(y-0.713)(y-1.112)}{(0.915-0.713)(0.915-1.112)} (7) + \frac{(y-0.713)(y-0.915)}{(1.112-0.713)(1.112-0.915)} x (10)$$

put $y = c_L = 1$

$x = \square$

Ans

(d)

$x = [1, 3, 7, 10];$

$y = [0.562, 0.713, 0.915, 1.112];$

$p = \text{polyfit}(x, y, 3);$

$x1 = [1.5, 2, 8];$

$y1 = \text{polyval}(p, x1)$

Quiz-0.2 → Sunday

1. Fixed point Iteration (one variable)
2. Newton's method (one & two variables)
3. Lagrangian interpolation.

$$A = \begin{bmatrix} & & & \end{bmatrix}$$

$$B = \begin{bmatrix} & \dot{} & \dot{} & \end{bmatrix}$$

$$x_{n+1} = \frac{1}{3}(2x_n^2 - 3y_n^2)$$

$$x(1) =$$

$$y(1) =$$

for $i = 1 : 6$

$$x(i+1) = (1/3) \left(2 \times \underline{x(i)^2} - 3 \times \underline{y(i)^2} \right)$$

$$y(i+1) = \dots$$

end

Newtonian Dividend Table :

x	1	3	6
$y=f(x)$	-3	11	62

a. Find $f(5)$ using Newtonian Dividend method.

b. Use two points to find the root. (% Find x when $y=0$)

Solⁿ:

(a)

x	y	Δy	$\Delta^2 y$
x_0 1	-3 a_0	$\frac{11 - (-3)}{3 - 1} = 7$ a_1	$\frac{17 - 7}{6 - 1} = 2$ a_2
x_1 3	11	$\frac{62 - 11}{6 - 3} = 17$	
x_2 6	62		

$$\therefore y = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$$

$$y = -3 + 7(x - 1) + 2(x - 1)(x - 3)$$

put $x = 5$,

$$y = -3 + 7(4) + 2(4)(2)$$

$$= -3 + 28 + 16$$

$$= 41 - 3$$

$$= 41$$

Ans

⑥

x	1	3
y	-3	11

	y	x	Δx
y_0	-3	① a_0	$\frac{3-1}{11+3} = \frac{2}{14}$
y_1	11	3	

$= \textcircled{\frac{1}{7}} a_1$

$$\therefore x = a_0 + a_1(y - y_0)$$

$$= 1 + \frac{1}{7}(y + 3)$$

$$x = 1 + \frac{1}{7}(0 + 3)$$

$$= 1 + \frac{3}{7} = \frac{10}{7} = 1.428 \quad \underline{\underline{\text{Ans}}}$$

Newtonian Dividend Table

question: To estimate freezing point (ফ্রিজিং পয়েন্ট), we need to determine Glaisher's constant, which is dependent on temp.

That $G = f(T)$

$T(^{\circ}C)$	20°	25°	27°	33°	35°
G	1.79	1.70	1.68	1.61	1.58

→ a. construct Dividend table.

→ b. Find $f(30)$ or, Find G at temperature $T = 30^{\circ}C$.

c. If $g(35^{\circ}C) = 1.58$, then find $g(T)$.

d. use two suitable points to find temperature when
 $G = 1.75$.

Soln:

(a)

	T	G	ΔG	$\Delta^2 G$	$\Delta^3 G$
T_0	20	1.79 $\rightarrow a_0$			
T_1	25	1.70	$\frac{1.70 - 1.79}{25 - 20} = -0.018 \rightarrow a_1$	$\frac{-0.01 + 0.018}{27 - 20} = 0.0011 \rightarrow a_2$	
			$\frac{1.68 - 1.70}{27 - 25} = -0.01$		$\frac{-0.016 - 0.0011}{33 - 20}$
T_2	27	1.68		$\frac{-0.117 + 0.01}{33 - 25} =$	$= -0.0013 \rightarrow a_3$
			$\frac{1.61 - 1.68}{33 - 27} = -0.117$	-0.016	
T_3	33	1.61			

$$\textcircled{b} \quad G = a_0 + a_1(T-T_0) + a_2(T-T_0)(T-T_1) + a_3(T-T_0)(T-T_1)(T-T_2)$$

$$\rightarrow f(T) = 1.79 - 0.018(T-20) + 0.0011(T-20)(T-25) - 0.0013(T-20)(T-25)(T-27)$$

$$\therefore G = f(30) = 1.79 - 0.018(10) + 0.0011(10)(5) - 0.0013(10)(5)(3)$$

$$= 1.47$$

$$\textcircled{c} \quad g(T) = f(T) + \boxed{a_4(T-T_0)(T-T_1)(T-T_2)(T-T_3)} \quad \text{---} \textcircled{1}$$

Given that, $g(35) = 1.58$

From (b) $\Rightarrow f(35) = 1.79 - 0.018(35-20) + 0.0011(35-20)(35-25) - 0.0013(15)(10)(8)$

$$= \boxed{x}$$

From (c) $T = 35$

$$1.58 = x + a_4(35-20)(35-25)(35-27)(35-33)$$

$$\therefore 1.58 - x = a_4(15)(10)(8)(2)$$

$$\therefore a_4 = \frac{1.58 - x}{30 \times 80} = \frac{1.58 - x}{2400} = \boxed{}$$

From *

$$g(T) = f(T) + \boxed{}(T-20)(T-25)(T-27)(T-33) \quad \text{Am}$$

①

T	20	25
G _r	1.79	1.70

g → Find T when G_r = 1.75

	G _r	T	ΔT
G ₀	1.79	20 → a ₀	$\frac{25-20}{1.70-1.79} = -55.56$ a ₁
G ₁	1.70	25	

$$T = a_0 + a_1 (G_r - G_0)$$

$$\therefore T = 20 - 55.56 (G_r - 1.79)$$

$$T = 20 - 55.56 (1.75 - 1.79)$$

$$\boxed{T = 22.2224^\circ\text{C}} \quad \text{Ans}$$

Given that,

x	-3	-1	3	7
y	8	6	1	-3

a. construct dividend table

b. Find $f(10)$. using the Newtonian Dividend method

c. Given that $g(10) = -8$. & $g(x)$ takes all the value of above table. Find $g(x)$

d. use two suitable points to find root. (% Find x when $y=0$)

e. " " " " / Linear method to find x when $y=7$.

soln:

a			Δy	$\Delta^2 y$	$\Delta^3 y$
	x	y			
x_0	-3	8	$\frac{6-8}{-1+3} = -1$	$\frac{-1.25+1}{3+3} = 0.0416$	
x_1	-1	6	$\frac{1-6}{3+1} = -1.25$	$\frac{-1+1.25}{7+1} = 0.03125$	$\frac{0.03125-0.0416}{7+3} = 0.00104$
x_2	3	1	$\frac{-3-1}{7-3} = -1$		
x_3	7	-3			

$$\textcircled{b} \quad f(x) = y = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$$

$$\therefore y = 8 - (x+3) + 0.0416(x+3)(x+1) + 0.00104(x+3)(x+1)(x-3)$$

at $x = 10$

$$y = 8 - 13 + 0.0416(13)(11) + 0.00104(13)(11)(7) = 1.98984$$

Ans

$\textcircled{c}$

x	-3	-1	3	7	10
y					-8

$$g(x) = f(x) + a_4(x+3)(x+1)(x-3)(x-7)$$

Given $g(10) = -8$

$$f(10) = 1.98984$$

$$-8 = 1.98984 + a_4(13)(11)(7)(3)$$

$$\therefore a_4 = \checkmark$$

$$\therefore g(x) = f(x) + \boxed{} (x+3)(x+1)(x-3)(x-7)$$

Ans

(d)

x	3	7
y	1	-3

	y	x	Δx
y_0	1	3	$\frac{4}{-4} = -1$
y_1	-3	7	

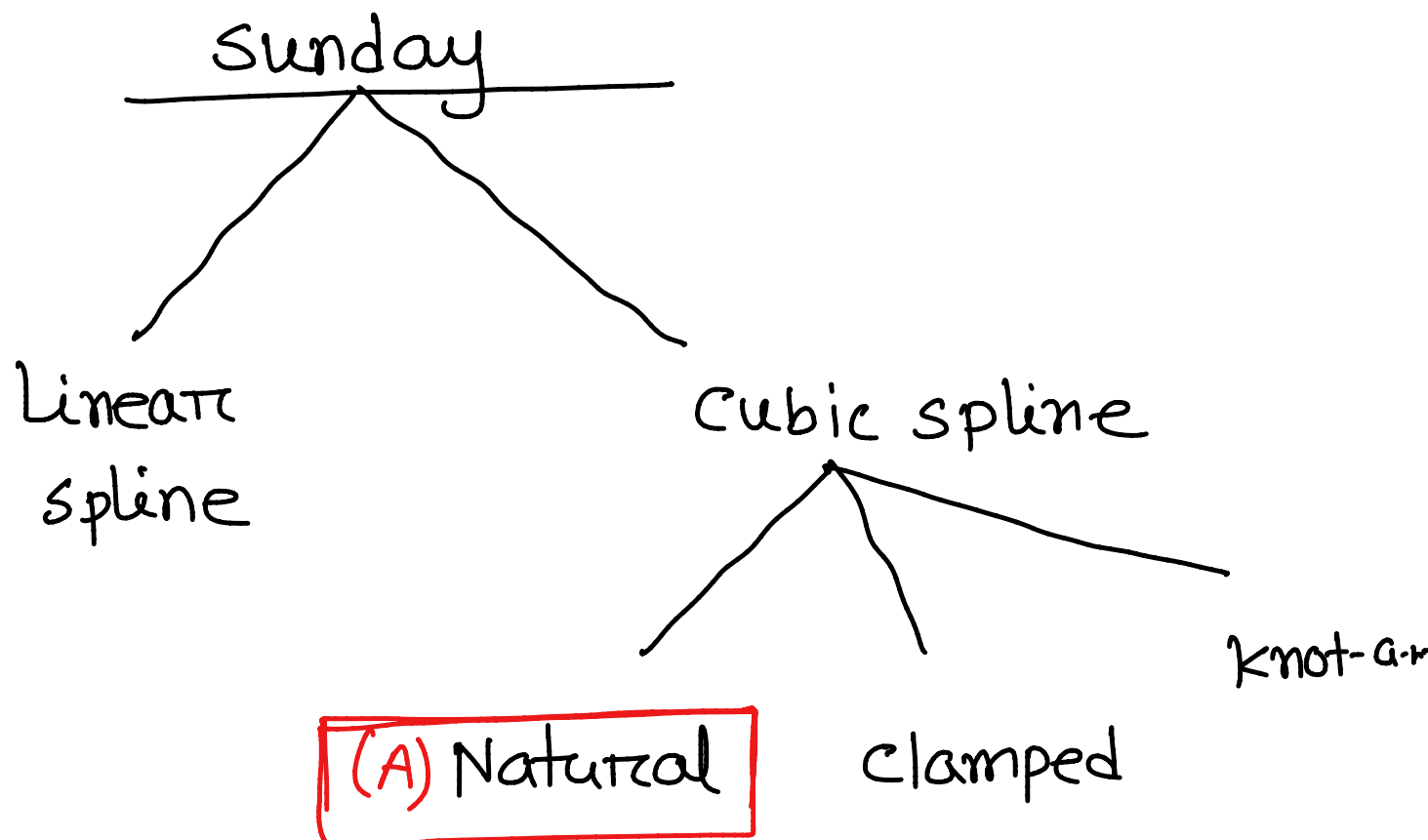
$$x = f(y) = 3 - (y - 1)$$

$$x = 3 - (0 - 1) \quad \% \text{ (root} \Rightarrow y = 0)$$

$$= 4$$

Ans

(e)



Spline

$$f(x) = \begin{cases} f_1(x) & , a \leq x \leq b \\ f_2(x) & , b \leq x \leq c \\ \vdots & \\ f_n(x) & , x_{n-1} \leq x \leq x_n \end{cases}$$

piecewise Function
↓
spline

Linear Spline:

$$f(x) = \begin{cases} ax+b, & x_1 \leq x \leq x_2 \\ bx-c, & x_2 \leq x \leq x_3 \end{cases}$$

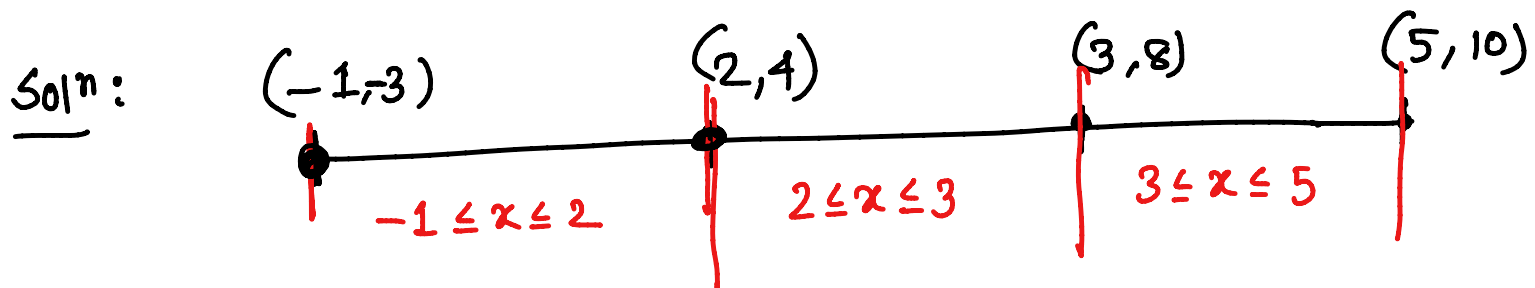
A straight line passing through (x_1, y_1) & (x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\Rightarrow y = y_1 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Q1. Construct linear spline that passing through $(-1, -3), (2, 4), (3, 8), (5, 10)$.

Hence find $f(0)$ & $f(4)$.



When $-1 \leq x \leq 2$, $(-1, -3)$ & $(2, 4)$

$$y = -3 + \frac{4+3}{2+1}(x+1) = -3 + \frac{7}{3}(x+1)$$

When $2 \leq x \leq 3$, $(2, 4)$ & $(3, 8)$

$$y = 4 + \frac{8-4}{3-2}(x-2) = 4 + 4(x-2)$$

When $3 \leq x \leq 5$, $(3, 8)$ & $(5, 10)$

$$y = 8 + \frac{10-8}{5-3}(x-3) = 8 + (x-3)$$

$$\therefore f(x) = \begin{cases} -3 + \frac{7}{3}(x+1), & -1 \leq x \leq 2 \\ 4 + 4(x-2), & 2 \leq x \leq 3 \\ 8 + (x-3), & 3 \leq x \leq 5 \end{cases} \rightarrow \text{linear spline}$$

Ans

$$f(0) = -3 + \frac{7}{3}(0+1) = -\frac{2}{3}$$

$$f(4) = 8 + (4-3) = 9 \quad \underline{\underline{\text{Ans}}}$$

construct linear spline that passing through $(-1, -8)$, $(2, 4)$ & $(5, 13)$.

Hence find $f(3.5)$.

Soln:

$$f(x) = \begin{cases} -8 + \frac{4+8}{2+1} (x+1), & -1 \leq x \leq 2 \\ 4 + \frac{13-4}{5-2} (x-2), & 2 \leq x \leq 5 \end{cases}$$

$$f(x) = \begin{cases} -8 + 4(x+1), & -1 \leq x \leq 2 \\ 4 + 3(x-2), & 2 \leq x \leq 5 \end{cases}$$

$$f(3.5) = 4 + 3(3.5-2) = 4 + 3 \times 1.5 = 4 + 4.5 = 8.5 \quad \underline{\underline{Ans}}$$

cubic spline

(1) Natural cubic spline:

at the boundary points: $f''(x) = 0$

at the interior points $f(x)$, $f'(x)$ & $f''(x)$

are continuous.

$$f(x) = \begin{cases} x^3 + 3, & 1 \leq x \leq 3 \\ x + 3, & 3 \leq x \leq 5 \end{cases}$$

$$9 + 3 \neq 6$$

not continuous.

boundary points: 1 & 5

interior points: 3

Given that, $f(x) = \begin{cases} A + 6(x+2) + B(x+2)^2 + C(x+2)^3, & -2 \leq x \leq 3 \\ 7 + D(x-3) - 12(x-3)^2 + E(x-3)^3, & 3 \leq x \leq 8. \end{cases}$

use the conditions for Natural cubic spline to find A, B, C, D, E & $f(0)$.

Soln:

$$f(x) = \begin{cases} A + 6(x+2) + B(x+2)^2 + C(x+2)^3, & -2 \leq x \leq 3 \\ 7 + D(x-3) - 12(x-3)^2 + E(x-3)^3, & 3 \leq x \leq 8 \end{cases}$$

$$f'(x) = \begin{cases} 6 + 2B(x+2) + 3C(x+2)^2, & -2 \leq x \leq 3 \\ D - 24(x-3) + 3E(x-3)^2, & 3 \leq x \leq 8 \end{cases}$$

$$f''(x) = \begin{cases} 2B + 6C(x+2), & -2 \leq x \leq 3 \\ -24 + 6E(x-3), & 3 \leq x \leq 8 \end{cases}$$

Boundary points: $-2, 8$

$$f''(-2) = 0$$

$$2B + 6C(-2+2) = 0$$

$$\therefore B = 0$$

$$f''(8) = 0$$

$$\Rightarrow -24 + 6E \times 5 = 0$$

$$\Rightarrow 30E = 24$$

$$\Rightarrow E = \frac{4}{5}$$

interior point $x=3$

$f''(x)$ is continuous at $x=3$

$$0 + 6C(5) = -24 + 0$$

$$\Rightarrow 30C = -24$$

$$\therefore C = -\frac{4}{5}$$

$f'(x)$ is continuous at $x=3$

$$6 + 0 + 3C(5)^2 = D + 0 + 0$$

$$\Rightarrow 6 + 3 \times \left(-\frac{4}{5}\right) \times 25 = D$$

$$D = -54$$

$f(x)$ is continuous at $x=3$

$$A + 6(5) + 0 + C(5)^3 = 7$$

$$\Rightarrow A + 30 - \frac{4}{5} \times 5^3 = 7$$

$$\Rightarrow A + 30 - 100 = 7$$

$$\Rightarrow A = 77$$

$$\therefore f(x) = \begin{cases} 77 + 6(x+2) - \frac{4}{5}(x+2)^3, & -2 \leq x \leq 3 \\ 7 - 54(x-3) - 12(x-3)^2 + \frac{4}{5}(x-3)^3, & 3 \leq x \leq 8 \end{cases}$$

$$f(0) = 77 + 6(2) - \frac{4}{5} \times (0+2)^3 = 77 + 12 - \frac{4}{5} \times 8$$

— 0 —

clamped cubic spline : (Set : B/c)

$f'(a) = m_1$ & $f'(b) = m_2$, a & b are boundary points.

interior point continuous.

knot-a - Not cubic spline ✓

MATLAB:

$x = [\quad]$

$y = [\quad]$

$f = \text{spline}(x, y)$

$x_{\text{new}} = [\quad]$

$y_{\text{new}} = \text{fnval}(f, x_{\text{new}})$

— 0 —